

Real-time prediction of intensive care unit patient acuity and therapy requirements using state-space modelling

Corresponding Author: Dr Parisa Rashidi

This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

This study proposes a Mamba-based deep learning model to predict the acuity state of patients in the ICU, changes in acuity states and the need for life-sustaining therapies. The model was trained using data from University of Florida Health and the eICU and MIMIC-IV datasets. A major strength of the study is that the model was validated on external, temporally shifted and prospectively collected data. The model performed significantly better than the traditional SOFA score.

Major comments:

- The motivation for using Mamba is not very clear. One would expect Mamba to shine when input sequences are too long for a Transformer to handle. However, possibly because inputs are restricted to four hour windows and patients with very long ICU stays are excluded, there doesn't seem to be a significant difference between APRICOT-T and APRICOT-M. Is it possible to better illustrate the advantages of Mamba, e.g. in terms of accuracy when longer input sequences are used, or alternatively in terms of computational requirements?
- It is unclear why positional encodings are used in APRICOT-M. Mamba generally doesn't require positional encodings, it would be good to present an ablation study illustrating their need.
- There are some additional limitations of the study that the authors could discuss. For example, confidence intervals are computed using bootstrapping, which provides uncertainty information related to the data sample but not related to model parameters. Also, using time-to-event analysis rather than binary classification could avoid some of the issues related to time window choice and dichotomy between analysis by step and by episode.
- There is little (or none?) discussion on the performance of the various baselines. In particular, CatBoost and GRU appear to be competitive on many tasks while likely requiring far less computational resources (an aspect of increasing relevance when developing and deploying machine learning models).
- Since the model was evaluated prospectively, it would have been interesting to report on clinicians' views on the use and clinical applicability of the model.

Minor comments:

Introduction

Line 49: I think the first paragraph of the introduction could be improved. E.g., what are the patient acuity categories? ICU patients likely have a potentially life-threatening condition independently of change in acuity.

71: I would clarify that outcomes in this study will be defined according to the paper by Ren et al., otherwise it's a bit unclear why you are introducing this paper in such detail.

100: how was the 2014-2017 time period selected for the development and external validation datasets, i.e. was this the time period covered by eICU and MIMIC or did you purposefully restrict the time period?

Methods

139: it would be nice to provide a histogram of the number of patients in each hospital as supplementary material, together with the cutoff of 800 patients per hospital for a hospital to be included in development set.

147 and 250: you mention that 10% of 'each' validation dataset was used for calibration. What do you mean by each? Isn't there one validation dataset?

220: what was the maximum sequence length for the model?

230: could you please clarify why temporal features were resampled to 20-minute resolution for GRU and vanilla Transformer?

238: how were hyperparameters selected, did you use a systematic approach?

245: top-k pooling is only mentioned in the description of fig. 3, could you please expand on what this is doing exactly?

282: are you using the bootstrapped samples and associated metrics for the Wilcoxon rank sum test? In this case I would reorder Section 2.9 starting with the description of bootstrapping.

Results:

320: I was a bit surprised to find out that, after all the various datasets had been nicely described in Methods, results for the temporal cohort were relegated to supplementary material. What is the reason for this? It doesn't seem that an extra column for the temporal cohort would take up a lot of extra space in the main article.

335: when you report accuracy metrics on the development dataset, do you mean the whole development dataset or just the 20% validation subset? Please clarify and motivate.

369: you mention using a 33% precision cutoff but in statistical analysis you mention the Youden index. Which one are you using? Also, you explain in the following paragraph why 33% is used but it could be explained already here.

388: I wonder how informative the number of early alerts in Table 5 is. Could you normalise it by the number of steps before the predicted outcome?

399: I assume that changing the time window of analysis will also change the balance between positive and negative classes, favouring the 4-h window in terms of AUROC. Would AUPRC be a better metric for comparison in Table 5?

430: Both examples of real-time evaluation do not appear to predict the change in acuity before MV and VP, arguably one of the most interesting applications of this model. I'd consider presenting an example where this is the case.

430: how did you compute predictive model uncertainty in figures 10 and 11?

Discussion

459: The authors state that AUROC for VP 'was higher than what is seen in the literature', but it appears comparable.

(Remarks on code availability)

Reviewer #2

(Remarks to the Author)

The paper discusses APRICOT-M, a state space-based neural network model designed to predict patient acuity in ICUs. It uses data from the preceding four hours, including vital signs, lab results, and patient characteristics, to predict outcomes like stability, instability, mortality, and the need for life-sustaining therapies in the next four hours. The model was trained on data from over 100,000 patients and validated on multiple datasets, showing superior performance compared to traditional methods like the SOFA score. APRICOT-M aims to assist clinicians in making timely interventions by providing real-time predictions without the need for data imputation.

Overall, the curation and harmonization of such diverse dataset require tremendous efforts, and the authors should be applauded for their dedication. I have a number clarifying questions/comments:

1) In the abstract the authors state "Early detection of deteriorating conditions can assist in more timely interventions and improved survival rates".

While there are some studies that provide evidence for such a claim in other hospital settings, is there any evidence for this statement within ICUs where patients are closely monitored and the nurse-to-patient ratios are typically higher?

2) The abstract states "it [the model] was validated ... prospectively on data from 215 patients (369 ICU admissions) from one 37 hospital in 2021-2023."

It seems like the Mamba model was first introduced by Gu and Dao (ref [20]) in 2023. This is confusing, since it appears that at the time the APRICOT-M model (based on Mamba) was being prospectively validated the Mamba architecture was not yet introduced in the literature. It appears that the APRICOT-M model was "retrospectively validated" on a dataset that was previously collected under a prospective data collection protocol. Please clarify.

3) It is also stated that "Data from UFH-P was used for prospective validation to evaluate the application of APRICOT-M to a real-time setting."

Please clarify if the model was applied to real-time data, and what real-time means in this context.

Similarly, Please expand on this sentence: "prospective model validation was performed on the data acquired using our clinical real-time platform [23] interfaced with the Epic® EHR system."

How was the model interfaced with the Epic EHR (HL7, FHIR, etc)?

Was the Epic EHR queried every 4 hours to make predictions for the subsequent 4 hours?

4) The Mamba Block in Figure 3 needs additional explanation and a description of what a "state-space model" is and why it is useful (preferably in an appendix). In particular, the model feels like a blackbox and it's not clear why it's providing an advantage over say vanilla transformers or other methods.

5) The authors state that "it is the first SSM model application to clinical data." A quick google scholar search of state-space models and application to clinical data reveals a number of prior studies:

* Yin Y, Chou CA. A novel switching state-space model for post-icu mortality prediction and survival analysis. IEEE Journal of Biomedical and Health Informatics. 2021 Mar 23;25(9):3587-95.

* Ghassemi M, Wu M, Hughes MC, Szolovits P, Doshi-Velez F. Predicting intervention onset in the ICU with switching state space models. AMIA Summits on Translational Science Proceedings. 2017;2017:82.

* Nemati S, Li-wei HL, Adams RP. Learning outcome-discriminative dynamics in multivariate physiological cohort time series. In 2013 35th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC) 2013 Jul 3 (pp. 7104-7107). IEEE.

Please explain how the proposed SSM model differs from prior works in this domain.

6) One of the challenges with prediction of rare events is having low positive predictive values (PPVs). In such settings, AUC as a metric tends to produce overly optimistic results. Nevertheless, the predictive model may still be clinically useful. I appreciate the two examples of patients discussed in section 3.7. It would be very insightful if the analysis could be repeated in a larger sample of 50 patients from the UFH-P cohort by at least three clinical adjudicators to determine the potential clinical utility of the proposed early warning system (were the alerts timely, accurate, actionable?).

(Remarks on code availability)

The project GitHub repo is well-organized, however the README file does not provide any information about software/hardware requirements/dependencies.

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

My comments were adequately addressed, I would like to commend the authors for their work.

(Remarks on code availability)

Reviewer #2

(Remarks to the Author)

Thank you for thoroughly responding to the question raised in my review. Overall, the manuscript has significantly improved. In particular, the clinical review findings by the three clinicians is insightful.

I only have a few minor comments for editorial review only (I don't expect a response):

1) Given your explanation of your evaluation method. The manner in which the term "[validated] prospectively" is used in the abstract may not be appropriate.

2) Given the relatively small sample size of 20 randomly selected cases for chart-review, it would be helpful to report the frequency of each outcome of interest (Hemodynamic instability, Respiratory distress, etc) and model recommendation and actual rate of each procedure (MV, VP, CRRT, etc).

3) The two references for evidence of impact of early detection on improved survival rates among ICU patients (namely Komorowski et al. and Shickel et al.) are not really valid, since they are not based on prospective clinical studies. You may want to tone down that claim.

(Remarks on code availability)

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Opening Statement

Firstly, the authors would like to sincerely thank the Editor and Reviewers for their time, thoughtful comments, and constructive feedback, which have greatly contributed to improving the quality of this manuscript. The following paragraphs are organized in the same sequence in which we received the reviews. We further extracted the comments (from Reviewer 1 and Reviewer 2) into bullet points which highlight the requested revision in each case. We have responded directly underneath each reviewer's comment. In case of modifications to the main/supplementary manuscript, we highlighted the page/line number in the response letter and the corresponding paragraphs in the main manuscript and/or supplemental material.

Reviewer 1

Major comments

Major comment 1. "The motivation for using Mamba is not very clear. One would expect Mamba to shine when input sequences are too long for a Transformer to handle. However, possibly because inputs are restricted to four hour windows and patients with very long ICU stays are excluded, there doesn't seem to be a significant difference between APRICOT-T and APRICOT-M. Is it possible to better illustrate the advantages of Mamba, e.g. in terms of accuracy when longer input sequences are used, or alternatively in terms of computational requirements?"

We thank the reviewer for their comment. We agree that the results presented do not show a significant difference between APRICOT-T and APRICOT-M. However, APRICOT-M (Mamba) does provide an advantage in terms of training/inference speed and has improved performance when using longer sequences. We have now included an analysis on performance using different sequence lengths for both APRICOT-T and APRICOT-M (Supplementary Figure S12) as well as a comparison in terms of inference speed (Supplementary Table S16).

For the purposes of this analysis, we expand the observation window from four hours to all data from the time of admission to the time of prediction. We believe that the results from these analyses demonstrate the robustness of Mamba for the purpose of this study as well as for future adaptations.

Major comment 2. "It is unclear why positional encodings are used in APRICOT-M. Mamba generally doesn't require positional encodings, it would be good to present an ablation study illustrating their need."

We thank the reviewer for their remark. Although positional encodings are not generally required by Mamba, we found that adding them resulted in a slight overall performance boost. We have added a table to the supplemental with the ablation study showing these results (Supplementary Table S15). We found that including positional encodings had higher

performance in 11 out of 36 tasks across all cohorts, while the model without positional encodings had higher performance in 9, with the remaining 16 tasks having the same performance for both conditions.

Major comment 3. " There are some additional limitations of the study that the authors could discuss. For example, confidence intervals are computed using bootstrapping, which provides uncertainty information related to the data sample but not related to model parameters. Also, using time-to-event analysis rather than binary classification could avoid some of the issues related to time window choice and dichotomy between analysis by step and by episode."

We thank the reviewer for pointing out these additional limitations. We agree that the confidence intervals computed via bootstrapping mainly reflect uncertainty related to the data sample, rather than uncertainty of the model parameters themselves. We also acknowledge the insightful suggestion regarding the use of a time-to-event analysis approach, which indeed could mitigate issues associated with the arbitrary choice of prediction time windows and reduce the dichotomy between step-level and episode-level analyses. We have incorporated these points explicitly into the limitations section of the Discussion to improve clarity and transparency (page 32, line 699).

Major comment 4. " There is little (or none?) discussion on the performance of the various baselines. In particular, CatBoost and GRU appear to be competitive on many tasks while likely requiring far less computational resources (an aspect of increasing relevance when developing and deploying machine learning models)."

We thank the reviewer for highlighting this important point. To provide a clearer picture of our model's overall competitiveness relative to baseline methods, we have compared the "win rate" of APRICOT-M against baselines to show the capability of our proposed method. Specifically, we define the win rate as the proportion of experimental comparisons (across multiple datasets and prediction tasks) in which our proposed method outperformed each baseline method individually.

Across all experiments conducted in this study, our method achieved a win rate of 61.1% (22 out of 36 comparisons) against CatBoost, and 66.7% (24 out of 36 comparisons) against GRU, 69.4% (25 out of 36) against transformer, and 61.1% (22 out of 36 comparisons) against APRICOT-T, demonstrating consistent advantages in predictive performance. We have added a paragraph discussing these results on (page 29, line 588).

Major comment 5. " Since the model was evaluated prospectively, it would have been interesting to report on clinicians' views on the use and clinical applicability of the model."

We thank the reviewer for their comment. We have successfully conducted an adjudication of the model by three clinicians to evaluate its clinical utility and applicability. We have explained

the methodology on (page 13, line 314), reported the results on (page 26, line 512), and discussed the findings on (page 30, line 648). We have added the adjudication standard operating procedure (SOP) along with the complete adjudication survey to the supplementary material (Supplementary Material, page 71-83).

Minor comments

Introduction

- Line 49: I think the first paragraph of the introduction could be improved. E.g., what are the patient acuity categories? ICU patients likely have a potentially life-threatening condition independently of change in acuity.
 - We thank the reviewer for their comment. We have modified the first paragraph to briefly discuss acuity categories and clarify that changes in acuity can lead to increased threat to life (i.e., their mortality risk would increase) rather than life-threatening conditions which might already be present at time of admission. (page 2, line 55)
- 71: I would clarify that outcomes in this study will be defined according to the paper by Ren et al., otherwise it's a bit unclear why you are introducing this paper in such detail.
 - We thank the reviewer for their comment. We have added a statement in the introduction to clarify that the outcomes in our study are defined following the approach used by Ren et al. (page 3, line 108)
- 100: how was the 2014-2017 time period selected for the development and external validation datasets, i.e. was this the time period covered by eICU and MIMIC or did you purposefully restrict the time period?
 - We thank the reviewer for their comment. The 2014-2017 time period was chosen to allow for a temporal validation with UFH data between 2018-2019. By training the model on earlier admissions and validating it on a subsequent cohort, we aimed to assess the model's ability to generalize across different time periods and evolving clinical practices.

Methods

- 139: it would be nice to provide a histogram of the number of patients in each hospital as supplementary material, together with the cutoff of 800 patients per hospital for a hospital to be included in development set.
 - We thank the reviewer for their suggestion. We have included a histogram showing the ICU admission counts in each hospital along with the cutoff for inclusion in the development set (Supplementary Figure S1). We would like to also clarify that the cutoff was set at 800 ICU admissions. We have modified the text and cohort figures accordingly for clarification (page 4, line 149; page 6, line 171).
- 147 and 250: you mention that 10% of 'each' validation dataset was used for calibration. What do you mean by each? Isn't there one validation dataset?

- We appreciate the reviewer's remark. We have clarified in both (page 4, line 162) and (page 11, line 278) that 10% of each validation cohort (development validation split, external, temporal, and prospective) was used for calibration.
- 220: what was the maximum sequence length for the model?
 - We thank the reviewer for their comment. The maximum sequence of the model was 512 after hyperparameter tuning using Optuna. We have clarified this point in (page 9, line 263) and added the optimal hyperparameters of the model (including sequence length) along with the hyperparameter search space to the supplemental (Supplementary Table S6)
- 230: could you please clarify why temporal features were resampled to 20-minute resolution for GRU and vanilla Transformer?
 - We appreciate the reviewer's remark. We resampled features to 20-minute resolution to keep a fixed sequence length across all features, given that GRU and vanilla Transformer required a fixed input dimension of (batch size, sequence length, number of features). This resolution allowed to keep some of the granularity of the temporal features while also avoiding excessive imputation for features with less frequency (e.g. lab, medications, scores). We have added a clarifying statement on (page 9, line 253).
- 238: how were hyperparameters selected, did you use a systematic approach?
 - We appreciate the reviewer's comment. We used the Optuna Python library for hyperparameter tuning. We have clarified this point on (page 9, line 263)
- 245: top-k pooling is only mentioned in the description of fig. 3, could you please expand on what this is doing exactly?
 - We appreciate the reviewer's remark. The output of the Mamba block is passed to a fully connected layer. To reduce the size and computational load of this layer, we apply top-k pooling, which selects the k most informative time steps from the sequence output. This reduces the Mamba output from a 3D tensor (batch size × sequence length × hidden dimension) to a 2D tensor (batch size × k × hidden dimension), making the fully connected layer more efficient. We have added a clarifying statement on (page 9, line 242).
- 282: are you using the bootstrapped samples and associated metrics for the Wilcoxon rank sum test? In this case I would reorder Section 2.9 starting with the description of bootstrapping.
 - We thank the reviewer for their comment. Yes, we are using the bootstrapped samples and associated metrics for the Wilcoxon rank sum test. We have reordered the section to clarify this point (page 12, line 310).

Results

- 320: I was a bit surprised to find out that, after all the various datasets had been nicely described in Methods, results for the temporal cohort were relegated to supplementary material. What is the reason for this? It doesn't seem that an extra column for the temporal cohort would take up a lot of extra space in the main article.
 - We thank the reviewer for their comment. Given that the prospective cohort (which is also from UFH) is also temporally shifted (2021-2023 compared to the

training period 2014-2017) we thought showing results of the temporal cohort on the main article could be redundant. Therefore, we moved them to the supplemental as additional validation for the model.

- 335: when you report accuracy metrics on the development dataset, do you mean the whole development dataset or just the 20% validation subset? Please clarify and motivate.
 - We thank the reviewer for their comment. All metrics and analyses reported for the development cohort are for the 20% validation subset. We report results on this subset rather than the entire development cohort to provide a more unbiased estimate of model performance and avoid overestimating results due to training data exposure. We have added a clarifying statement on (page 15, line 369).
- 369: you mention using a 33% precision cutoff but in statistical analysis you mention the Youden index. Which one are you using? Also, you explain in the following paragraph why 33% is used but it could be explained already here.
 - We thank the reviewer for their comment. We used the Youden index to obtain the optimal threshold and calculate metrics at the step level for all models and cohorts (reported on Supplemental Table S6), while results for episode-level (and resulting step-level) are calculated for different episode precision thresholds. We have clarified this point on (page 15, line 369). We have also modified the paragraph to explain why 33% was used (page 18, line 427).
- 388: I wonder how informative the number of early alerts in Table 5 is. Could you normalise it by the number of steps before the predicted outcome?
 - We thank the reviewer for this thoughtful suggestion. We would like to clarify that the information about the number of steps before the predicted outcome is already implicitly captured in the “Earliest prediction” column of Table 5, which reports the median (and interquartile range) of the earliest time at which an alert was issued relative to the eventual event. A longer earliest prediction time indicates that the model is able to provide alerts farther in advance, spanning more steps before the outcome. In addition, we believe the “Number of early alerts” column provides complementary information that is critical for understanding model behavior: it reflects how frequently the model issues early warnings per episode. A low number of early alerts indicates that the model remains conservative, avoiding frequent alarms and thus potentially reducing alarm fatigue, while still providing actionable early warnings. Together, the “Earliest prediction” and “Number of early alerts” offer a more complete picture of both the timeliness and burden of alerts generated by the system.
- 399: I assume that changing the time window of analysis will also change the balance between positive and negative classes, favouring the 4-h window in terms of AUROC. Would AUPRC be a better metric for comparison in Table 5?
 - We appreciate the reviewer’s suggestion. Although the lower class balance might favour the 4-h window in terms of AUROC, the larger time windows would be favoured by the AUPRC due to the higher class prevalence.

- 430: Both examples of real-time evaluation do not appear to predict the change in acuity before MV and VP, arguably one of the most interesting applications of this model. I'd consider presenting an example where this is the case.
 - We appreciate the reviewer's suggestion. We have added an additional example from the prospective cohort to show how APRICOT-M predicts the change in acuity early on in the admission before instability onset (page 23, line 502).
- 430: how did you compute predictive model uncertainty in figures 10 and 11?
 - We appreciate the reviewer's remark. Model uncertainty was computed by running the model 100 times at each time step and taking the lowest and highest outputs (predicted probabilities) as the range of uncertainty. We have added clarifying statements to the examples figure captions (Fig. 10, 11, and 12).

Discussion

- 459: The authors state that AUROC for VP 'was higher than what is seen in the literature', but it appears comparable.
 - We appreciate the reviewer's remark. We have modified the sentence to clarify (page 28, line 573). VP prediction on the development set was comparable (0.81 CI: 0.80-0.82) to what is seen in the literature (AUROC 0.80-0.81) while in the prospective set it was slightly higher (0.83 CI: 0.78-0.87).

Reviewer 2

1) In the abstract the authors state "Early detection of deteriorating conditions can assist in more timely interventions and improved survival rates".

While there are some studies that provide evidence for such a claim in other hospital settings, is there any evidence for this statement within ICUs where patients are closely monitored and the nurse-to-patient ratios are typically higher?

We thank the reviewer for their thoughtful comment. We agree that ICU patients are closely monitored with higher nurse-to-patient ratios. However, multiple studies have demonstrated that even in the ICU, early detection of subtle physiological deterioration—beyond standard monitoring—can facilitate more timely interventions and is associated with improved clinical outcomes. For instance, Komorowski et al. (2018, Nature Medicine) showed that AI-driven early warning systems in ICUs can recommend interventions that significantly reduce mortality. We have added references to clarify this point. Additionally, Shickel et al. (2018, DeepSOFA) proposed a deep learning-based severity scoring method that continuously evaluates ICU patients' conditions and more accurately predicts in-hospital mortality compared to traditional SOFA scores, enabling earlier identification of high-risk patients. We have added these references to clarify this point in the introduction section (page 2, line 59).

2) The abstract states "it [the model] was validated ... prospectively on data from 215 patients (369 ICU admissions) from one 37 hospital in 2021-2023."

It seems like the Mamba model was first introduced by Gu and Dao (ref [20]) in 2023. This is confusing, since it appears that at the time the APRICOT-M model (based on Mamba) was being prospectively validated the Mamba architecture was not yet introduced in the literature. It appears that the APRICOT-M model was “retrospectively validated” on a dataset that was previously collected under a prospective data collection protocol. Please clarify.

We thank the reviewer for their thoughtful comment. We validated the model retrospectively on data collected prospectively to simulate a prospective validation of the model. We have clarified this point on (page 5, line 168).

3) It is also stated that “Data from UFH-P was used for prospective validation to evaluate the application of APRICOT-M to a real-time setting.”

Please clarify if the model was applied to real-time data, and what real-time means in this context.

Similarly, Please expand on this sentence: “prospective model validation was performed on the data acquired using our clinical real-time platform [23] interfaced with the Epic® EHR system.”

How was the model interfaced with the Epic EHR (HL7, FHIR, etc)?

Was the Epic EHR queried every 4 hours to make predictions for the subsequent 4 hours?

We thank the reviewer for their thoughtful comment. In this context, real-time means the data was acquired in real-time through an automated pipeline querying the EPIC EHR database once per day during the collection period (2021-2023). The model was not interfaced with the EPIC EHR. Rather, we developed a separate interface to show model predictions in real-time. However, a validation of the real-time predictions will be performed on a subsequent study. We have added a figure of the web application interface for the real-time deployment of APRICOT-M on the supplemental (Supplementary Figure S16).

4) The Mamba Block in Figure 3 needs additional explanation and a description of what a “state-space model” is and why it is useful (preferably in an appendix). In particular, the model feels like a blackbox and it’s not clear why it’s providing an advantage over say vanilla transformers or other methods.

We thank the reviewer for their suggestion. We agree that additional explanation of the Mamba Block and state-space models would improve the clarity of our manuscript. In response, we have added a detailed description to the supplemental (Supplementary Material, pages 1-2). Specifically, we explain the structure of the Mamba Block and introduce the concept of state-space models. We also discuss why they provide advantages compared to traditional architectures like vanilla transformers in the discussion section (page 29, line 592).

5) The authors state that “it is the first SSM model application to clinical data.” A quick google scholar search of state-space models and application to clinical data reveals a number of prior studies:

* Yin Y, Chou CA. A novel switching state-space model for post-icu mortality prediction and survival analysis. IEEE Journal of Biomedical and Health Informatics. 2021 Mar 23;25(9):3587-95.

* Ghassemi M, Wu M, Hughes MC, Szolovits P, Doshi-Velez F. Predicting intervention onset in the ICU with switching state space models. AMIA Summits on Translational Science Proceedings. 2017;2017:82.

* Nemati S, Li-wei HL, Adams RP. Learning outcome-discriminative dynamics in multivariate physiological cohort time series. In 2013 35th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC) 2013 Jul 3 (pp. 7104-7107). IEEE.

Please explain how the proposed SSM model differs from prior works in this domain.

We thank the reviewer for the opportunity to address this concern. The proposed model was, to the best of our knowledge, the first application of Mamba (from A. Gu and T. Dao, “Mamba: Linear-Time Sequence Modeling with Selective State Spaces,” Dec. 01, 2023) to clinical data at the time of submission. However, more Mamba-based models have been applied to clinical data during the review period. We have modified this statement on (page 4, line 125) to clarify it is among the first applications of Mamba for clinical data. We have also cited some works using Mamba for clinical/EHR data and modified the text to clarify the specific use of Mamba-based models (page 3, line 95).

6) One of the challenges with prediction of rare events is having low positive predictive values (PPVs). In such settings, AUC as a metric tends to produce overly optimistic results. Nevertheless, the predictive model may still be clinically useful. I appreciate the two examples of patients discussed in section 3.7. It would be very insightful if the analysis could be repeated in a larger sample of 50 patients from the UFH-P cohort by at least three clinical adjudicators to determine the potential clinical utility of the proposed early warning system (were the alerts timely, accurate, actionable?).

We thank the reviewer for their valuable feedback. We have successfully conducted an adjudication with three clinicians to evaluate the clinical utility of APRICOT-M. Although we could not complete the analysis on 50 patients, we believe the results presented on 20 patients provide meaningful insight into the model’s clinical application. We have added a new subsection to the Methods section explaining the protocol (page 13, line 314), a new Results subsection with the findings (page 26, line 512), and a paragraph to the Discussion explaining the results (page 30, line 648).

We have added the adjudication standard operating procedure (SOP) along with the complete adjudication survey to the supplementary material (Supplementary Material, page 71-83).

The project GitHub repo is well-organized, however the README file does not provide any information about software/hardware requirements/dependencies.

We thank the reviewer for their remark. We have updated the README file on the github repo to include software/hardware requirements/dependencies
(<https://github.com/iheallab/apricotM>)

Opening Statement

Firstly, the authors would like to sincerely thank the Editor and Reviewers for their time, thoughtful comments, and constructive feedback, which have greatly contributed to improving the quality of this manuscript. The following paragraphs are organized in the same sequence in which we received the reviews. We further extracted the comments (from Reviewer 1 and Reviewer 2) into bullet points which highlight the requested revision in each case. We have responded directly underneath each reviewer's comment. In case of modifications to the main/supplementary manuscript, we highlighted the page number in the response letter.

Reviewer 1

My comments were adequately addressed, I would like to commend the authors for their work.

We would like to thank the reviewer for their feedback which improved the manuscript significantly.

Reviewer 2

Thank you for thoroughly responding to the question raised in my review. Overall, the manuscript has significantly improved. In particular, the clinical review findings by the three clinicians is insightful.

We would like to thank the reviewer for their feedback which improved the manuscript significantly.

I only have a few minor comments for editorial review only (I don't expect a response):

1) Given your explanation of your evaluation method. The manner in which the term "[validated] prospectively" is used in the abstract may not be appropriate.

We thank the reviewer for their comment. We have updated the abstract to substitute the term "validated prospectively" to "validated on real-time data" (page 1).

2) Given the relatively small sample size of 20 randomly selected cases for chart-review, it would helpful to report the frequency of each outcome of interest (Hemodynamic instability, Respiratory distress, etc) and model recommendation and actual rate of each procedure (MV, VP, CRRT, etc).

We appreciate the reviewer's suggestion. While we agree that reporting the frequency of each clinical outcome (Hemodynamic instability, Respiratory distress, etc) would provide additional context, this information is not readily available in structured form and would require extensive manual chart review to extract from unstructured clinical notes. Given the scope and timeline

of the current study, we were unable to perform a detailed analysis of each outcome category. However, we have included summary statistics of model-predicted and actual administration rates for life-sustaining therapies (MV, VP, CRRT, and BT) where structured data were available (Supplementary Figure S24, page 80 of Supplementary Information).

3) The two references for evidence of impact of early detection on improved survival rates among ICU patients (namely Komorowski et al. and Shickel et al.) are not really valid, since they are not based on prospective clinical studies. You may want to tone down that claim.

We thank the reviewer for their feedback. We have modified the sentence accordingly to tone down the claim (page 1).